Shamyo Brotee

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sbrotee63.github.io sbrotee63 Shamyo Brotee

Research Interests

Diffusion Models, Flow Matching, Molecular Graph Generation, AI for Science, Reinforcement Learning

Education

Texas State University — San Marcos, Texas
MS (Thesis) in Computer Science — Aug 2025 – Present

- GPA: 4.0 / 4.0 (One semester remaining)
- Thesis: Application of discrete-state continuous-time flow matching for size-adaptive molecular graph generation

University of Dhaka — Dhaka, Bangladesh
BS in Computer Science and Engineering — Jan 2018 – Jan 2023

- Thesis: Safe navigation of autonomous vehicles using reinforcement learning

Research Experience

Texas State University — San Marcos, Texas
Graduate Research Assistant (**Supervisor: Dr. Chul-Ho Lee**) — Aug 2025 - Present

- Working on a method of applying *diffusion models* and *flow matching* in generating size-adaptive property-conditioned *molecular graph structures*

Green Networking Research (GNR) Group, University of Dhaka — Dhaka, Bangladesh
Undergraduate Research Assistant (Supervisor: Dr. Md. Abdur Razzaque) — Jan 2022 - Jun 2024

- Explored several *reinforcement learning algorithms* and their applications in projects related to *networking* and *cyber-physical systems*
- Developed and published a hybrid framework integrating *mean-shift clustering* and *multi-agent deep reinforcement learning* for *minimal-collision path-planning* of *autonomous vehicles* in *obstructed environments*

Publications

Optimizing UAV-UGV Coalition Operations: A Hybrid Clustering and Multi-Agent Reinforcement Learning Approach for Path Planning in Obstructed Environment — Jul 1, 2024

Journal: **Elsevier Ad Hoc Networks** (Q1 Journal)

Authors: ***Shamyo Brotee***, Farhan Kabir, Md. Abdur Razzaque, Palash Roy, Md. Mamun-Or-Rashid, Md. Rafiul Hassan, Mohammad Mehedi Hassan

DOI: 10.1016/j.adhoc.2024.103519

Professional Experience

Enosis Solutions — Dhaka, Bangladesh
Software Engineer — Feb 2023 – Jul 2024

- Worked in the *data analytics team* of a US-based healthcare practice system
- Engaged in *system design*, *development*, *database query optimization* and *mainte-*

nance of the healthcare data infrastructure
- Involved in *cleaning and processing* patient data, *building BI analytics dashboards*

Teaching Experience

Shanto-Mariam University of Creative Technology — Dhaka, Bangladesh
Faculty Lecturer in Computer Science and Engineering — Mar 2025 – Aug 2025

- Instructing undergraduate courses on *Data Structures and Algorithms*, *Object- Oriented Programming* and *Machine Learning*

Awards, Grants and Scholarships

Texas State Ingram Hall Makerspace Digital Twin Project Awards — 2026
- Co-authored and secured a $7000 grant for our team's research project *SafeMake: A Digital Twin based Framework for Makerspace Safety and Knowledge Retrieval*

Texas State Graduate Merit Fellowship — 2025-2026
- Received a $2573 fellowship award for 2025-2026 Academic Year

Code Samurai — 2022
- Secured 4th place in an inter-university hackathon arranged by the embassy of Japan in Bangladesh

Standardized Tests

GRE — Sep 18, 2023
Total Score: 325/340 (Quant: 169/170, Verbal: 156/170, AWA: 5.0/6.0)

TOEFL — Oct 28, 2023
Total Score: 108/120 (Reading: 30/30, Listening: 26/30, Speaking: 27/30, Writing: 25/30)

Projects

Tag — 2022
- A *Multi-Agent Deep Reinforcement Learning* based *predator-prey game*, using *C#* and *Unity game engine*

CSEDU Gossip Bot — 2020
- An *educational chatbot* for academic queries regarding our department, using *Python*

Donate+ — 2019
- An *Android app* that caters to people in *urgent need of blood donors and recipients*, using *Java*

The Lone Ranger — 2018
- An *action-platform shooter game*, using *C++* and *iGraphics library*

Technical Skills

Language: Python, C++, C#, C, Java, JavaScript
Machine Learning: PyTorch, Scikit-Learn, Panda, Numpy, Matplotlib, Seaborn
Database: Microsoft SQL Server, MySQL, Microsoft Azure, Azure Data Studio, SSMS
Miscellaneous: Git, .NET, Microsoft Power BI, System Design